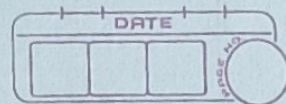


Assignment



Assignment 1: Study Hours vs Exam Marks.

Scenario: A teacher wants to determine whether students who study for more hours tend to score higher marks.

student	study Hours (X)	Marks (Y)
A	2	45
B	4	50
C	6	65
D	8	80
E	10	90

Questions:

1. calculate the mean of X.

→

$$\bar{X} = \frac{2+4+6+8+10}{5} = \frac{30}{5} = 6$$

2. calculate the mean of Y.

→

$$\bar{Y} = \frac{45+50+65+80+90}{5} = \frac{330}{5} = 66$$

3. Find the deviations $(X - \bar{X})$ and $(Y - \bar{Y})$.

→

X	Y	$X - \bar{X}$	$Y - \bar{Y}$
2	45	-4	-21
4	50	-2	-16
6	65	0	-1
8	80	2	14
10	90	4	24

4. Calculate the Product $(X - \bar{X})(Y - \bar{Y})$ for each observations.

$X - \bar{X}$	$Y - \bar{Y}$	Product
-4	-21	84
-2	-16	32
0	-1	0
2	14	28
4	24	96

$$\sum (X - \bar{X})(Y - \bar{Y}) = 84 + 32 + 0 + 28 + 96 = 240$$

5. Calculate the sample Covariance.

Formula: $\text{Cov}(X, Y) = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{n - 1}$

$$\text{Cov}(X, Y) = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{n - 1}$$

$$= \frac{240}{5 - 1} = \frac{240}{4} = 60$$

6. Interpret whether the Covariance indicates a Positive, negative, or no relationship.

The Covariance is Positive (60).

7. Verify your answer using Python (numpy.cov).

Assignment 2: Advertising Budget vs Sales.

Scenario: A Company wants to determine whether increasing its advertising budget leads to higher sales.

Month	Advertising Budget (₹ Thousand) (X)	Sales (₹ Thousand) (Y)
Jan	10	120
Feb	15	150
Mar	20	180
Apr	25	210
May	30	240

Questions:

1. calculate the mean of X.

→

$$\bar{X} = \frac{10 + 15 + 20 + 25 + 30}{5} = \frac{100}{5} = 20$$

2. calculate the mean of Y.

→

$$\bar{Y} = \frac{120 + 150 + 180 + 210 + 240}{5} = \frac{900}{5} = 180$$

3. Find the deviations $(X - \bar{X})$ and $(Y - \bar{Y})$.

→

X	Y	$X - \bar{X}$	$Y - \bar{Y}$
10	120	-10	-60
15	150	-5	-30
20	180	0	0
25	210	5	30
30	240	10	60

4. Calculate the Product $(x - \bar{x})(y - \bar{y})$ for each observations.

→

$x - 20$	$y - 180$	Product
-10	-60	600
-5	-30	150
0	0	0
5	30	150
10	60	600

$$\sum (x - \bar{x})(y - \bar{y}) = 600 + 150 + 0 + 150 + 600 = 1500$$

5. Calculate the sample Covariance

→

$$\text{Cov}(x, y) = \frac{\sum (x - \bar{x})(y - \bar{y})}{n - 1}$$

$$= \frac{1500}{5 - 1} = \frac{1500}{4} = 375$$

6. Interpret whether the Covariance indicates a Positive, Negative or no relationship.

→

The Covariance is Positive (+375)

7. Verify your answer using Python (numpy.cov)

$y - 180$	$x - 20$	y	x
-60	-10	120	10
-30	-5	150	15
0	0	180	20
30	5	210	25
60	10	240	30